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Course/Section: CPE212-CPE31S1	Date Submitted: November 9, 2025
Instructor: Engr. Robin Valenzuela	Semester and SY: 2nd Sem 2025-2026

Activity 1: Configure Network using Virtual Machines

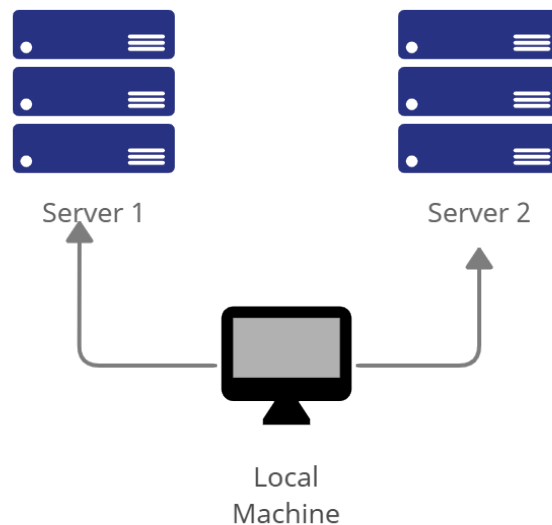
1. Objectives:

- 1.1. Create and configure Virtual Machines in Microsoft Azure or VirtualBox
- 1.2. Set-up a Virtual Network and Test Connectivity of VMs

2. Discussion:

Network Topology:

Assume that you have created the following network topology in Virtual Machines, *provide screenshots for each task*. (Note: *it is assumed that you have the prior knowledge of cloning and creating snapshots in a virtual machine*).

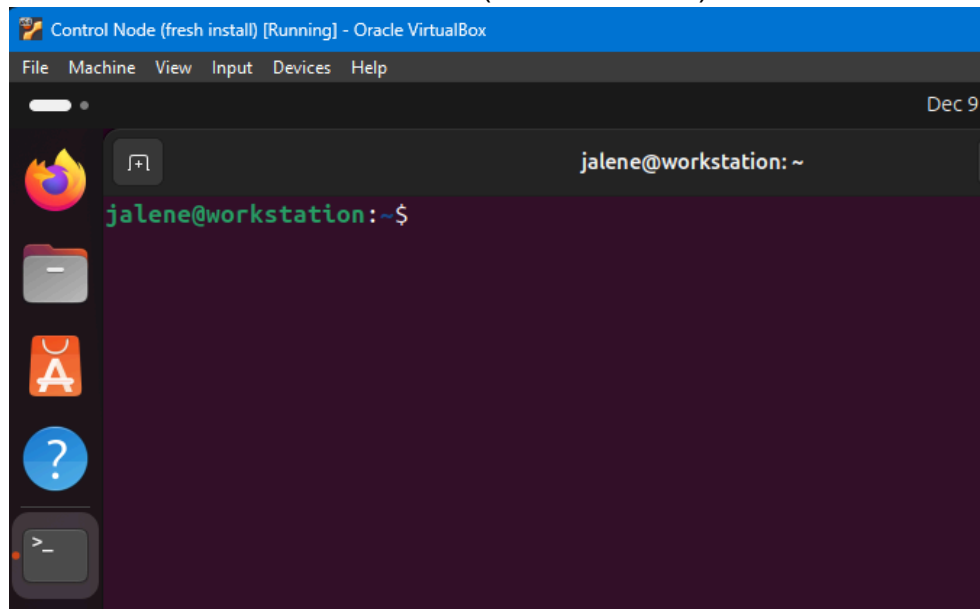


Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

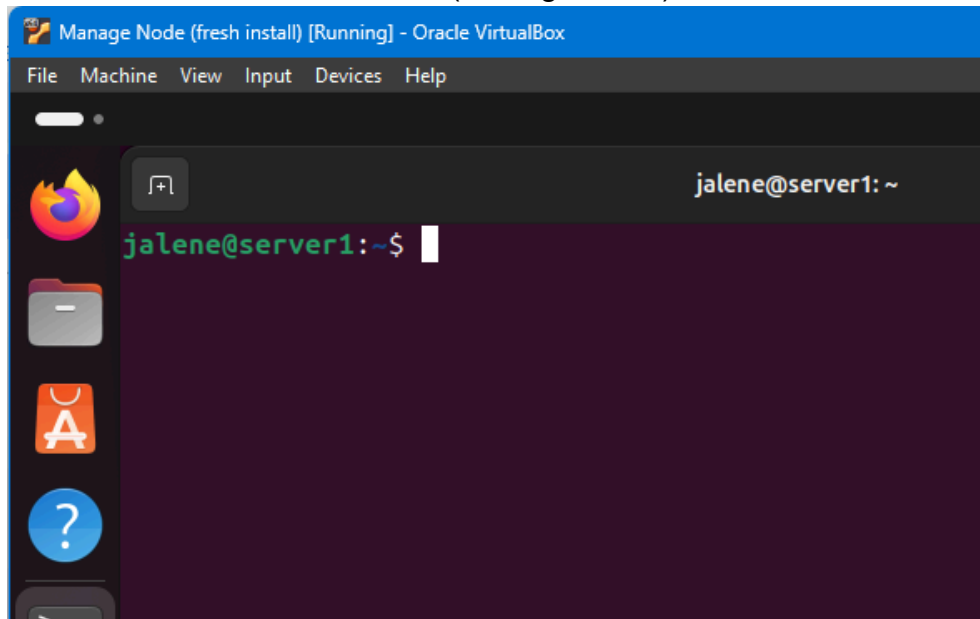
1. Change the hostname using the command *sudo nano /etc/hostname*
 - 1.1 Use server1 for Server 1
 - 1.2 Use server2 for Server 2
 - 1.3 Use workstation for the Local Machine
2. Edit the hosts using the command *sudo nano /etc/hosts*. Edit the second line.

- 2.1 Type 127.0.0.1 server 1 for Server 1
- 2.2 Type 127.0.0.1 server 2 for Server 2
- 2.3 Type 127.0.0.1 workstation for the Local Machine

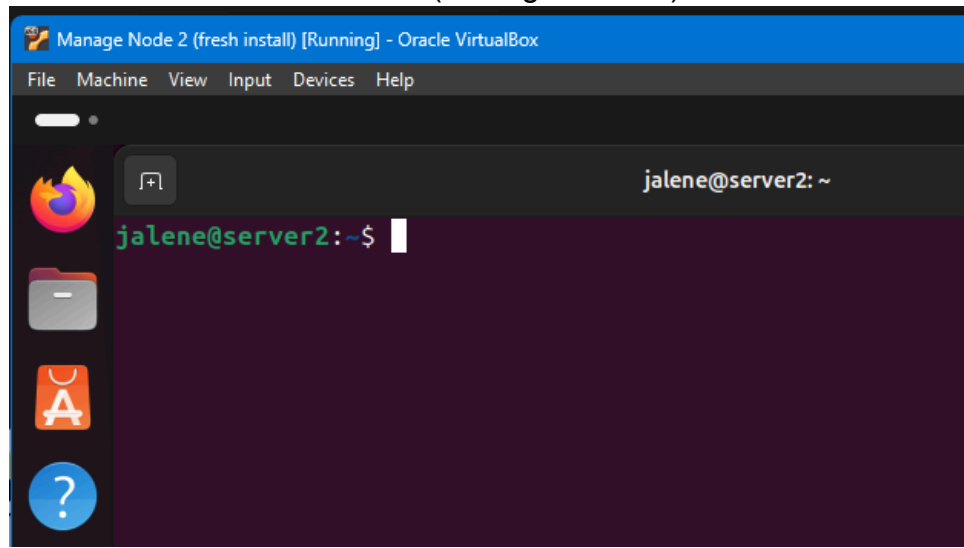
Control Node (Local Machine)



Server 1 (Manage Node)



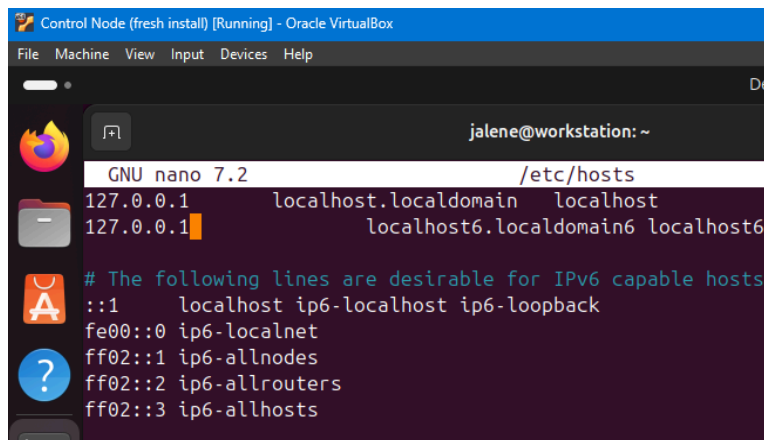
Server 2 (Manage Node 2)



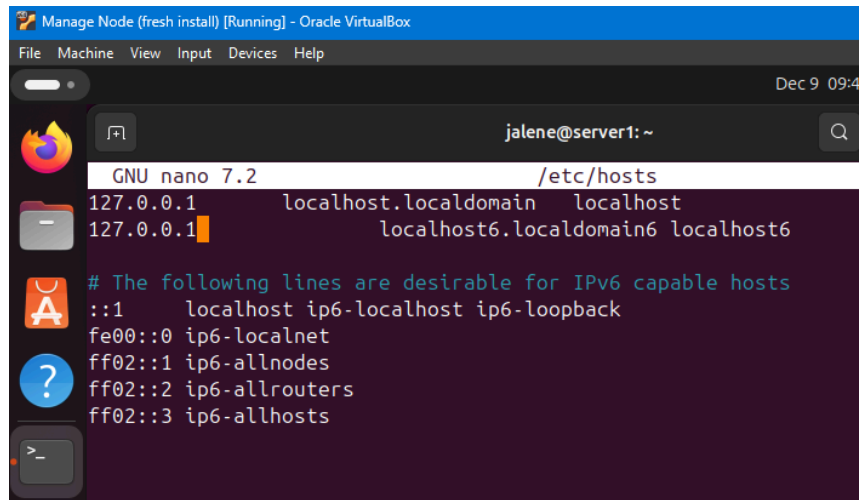
Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the command *sudo apt update* and *sudo apt upgrade* respectively.
2. Install the SSH server using the command *sudo apt install openssh-server*.
3. Verify if the SSH service has started by issuing the following commands:
 - 3.1 *sudo service ssh start*
 - 3.2 *sudo systemctl status ssh*
4. Configure the firewall to all port 22 by issuing the following commands:
 - 4.1 *sudo ufw allow ssh*
 - 4.2 *sudo ufw enable*
 - 4.3 *sudo ufw status*

Local Machine



Server 1

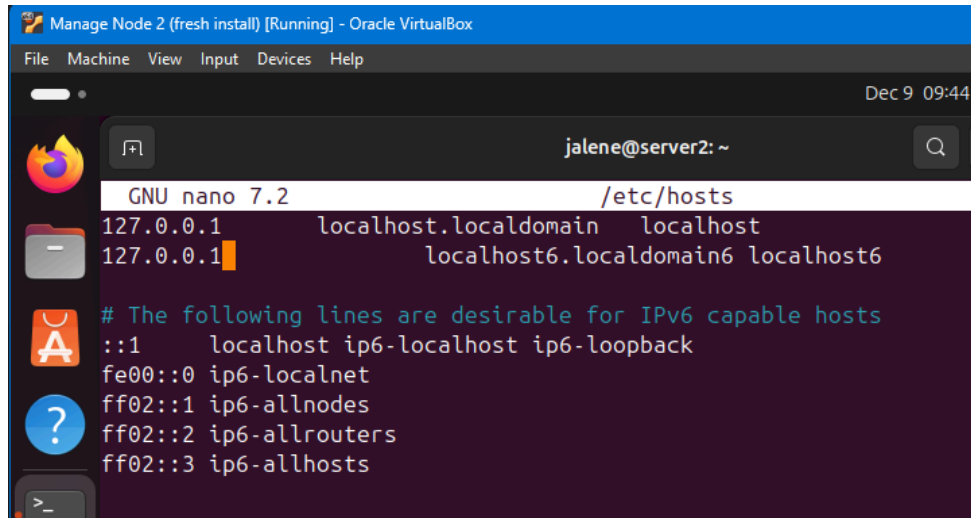


The screenshot shows a terminal window titled "Manage Node (fresh install) [Running] - Oracle VirtualBox". The user is logged in as "jalene@server1: ~". The terminal displays the contents of the `/etc/hosts` file, which is being edited with GNU nano 7.2. The file contains the following entries:

```
127.0.0.1 localhost.localdomain localhost
127.0.0.1 localhost6.localdomain6 localhost6

# The following lines are desirable for IPv6 capable hosts
::1 localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

Server 2



The screenshot shows a terminal window titled "Manage Node 2 (fresh install) [Running] - Oracle VirtualBox". The user is logged in as "jalene@server2: ~". The terminal displays the contents of the `/etc/hosts` file, which is being edited with GNU nano 7.2. The file contains the following entries:

```
127.0.0.1 localhost.localdomain localhost
127.0.0.1 localhost6.localdomain6 localhost6

# The following lines are desirable for IPv6 capable hosts
::1 localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```



Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command *ifconfig* and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.

1.1 Local Machine IP address: 192.168.56.113

1.2 Server 1 IP address: 192.168.56.114

1.3 Server 2 IP address: 192.168.56.115

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: ☐ Successful ☐ Not Successful

2.2 Connectivity test for Virtual Local Machine 1 to Server 2: ☐ Successful ☐ Not Successful

2.3 Connectivity test for Server 1 to Server 2: ☐ Successful ☐ Not Successful

The screenshot displays three overlapping terminal windows from Oracle VM VirtualBox, each running a Linux distribution named 'jalene'. The windows show the output of the 'ifconfig' command for different network interfaces.

- Left Window (jalene@workstation:~):** Shows configuration for interface 'enp0s3' with IP 192.168.56.113 and 'lo' with IP 127.0.0.1. It also shows details for 'eth0' and 'obal'.
- Middle Window (jalene@server1:~):** Shows configuration for interface 'enp0s3' with IP 192.168.56.114 and 'lo' with IP 127.0.0.1. It also shows details for 'eth0' and 'obal'.
- Right Window (jalene@server2:~):** Shows configuration for interface 'enp0s3' with IP 192.168.56.115 and 'lo' with IP 127.0.0.1. It also shows details for 'eth0' and 'obal'.

The terminals show various network statistics such as RX/TX packets, bytes, and errors, indicating that the network interfaces are up and running.

```
Control Node (fresh install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 9 10:05
jalene@workstation: ~
jalene@workstation: ~
jalene@workstation:~$ ping 192.168.56.114
PING 192.168.56.114 (192.168.56.114) 56(84) bytes of data.
64 bytes from 192.168.56.114: icmp_seq=1 ttl=64 time=0.776 ms
64 bytes from 192.168.56.114: icmp_seq=2 ttl=64 time=0.695 ms
64 bytes from 192.168.56.114: icmp_seq=3 ttl=64 time=3.14 ms
64 bytes from 192.168.56.114: icmp_seq=4 ttl=64 time=0.612 ms
64 bytes from 192.168.56.114: icmp_seq=5 ttl=64 time=0.316 ms
64 bytes from 192.168.56.114: icmp_seq=6 ttl=64 time=0.661 ms
^Z
[1]+  Stopped                  ping 192.168.56.114
jalene@workstation:~$ ping 192.168.56.115
PING 192.168.56.115 (192.168.56.115) 56(84) bytes of data.
64 bytes from 192.168.56.115: icmp_seq=1 ttl=64 time=1.18 ms
64 bytes from 192.168.56.115: icmp_seq=2 ttl=64 time=0.552 ms
64 bytes from 192.168.56.115: icmp_seq=3 ttl=64 time=0.659 ms
64 bytes from 192.168.56.115: icmp_seq=4 ttl=64 time=0.518 ms
64 bytes from 192.168.56.115: icmp_seq=5 ttl=64 time=0.609 ms
^Z
[2]+  Stopped                  ping 192.168.56.115
jalene@workstation:~$
```

```
Manage Node (fresh install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 9 10:06
jalene@server1: ~
jalene@server1: ~
jalene@server1:~$ ping 192.168.56.115
PING 192.168.56.115 (192.168.56.115) 56(84) bytes of data.
64 bytes from 192.168.56.115: icmp_seq=1 ttl=64 time=0.690 ms
64 bytes from 192.168.56.115: icmp_seq=2 ttl=64 time=1.34 ms
64 bytes from 192.168.56.115: icmp_seq=3 ttl=64 time=0.339 ms
64 bytes from 192.168.56.115: icmp_seq=4 ttl=64 time=0.729 ms
64 bytes from 192.168.56.115: icmp_seq=5 ttl=64 time=0.449 ms
^Z
[1]+  Stopped                  ping 192.168.56.115
jalene@server1:~$
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

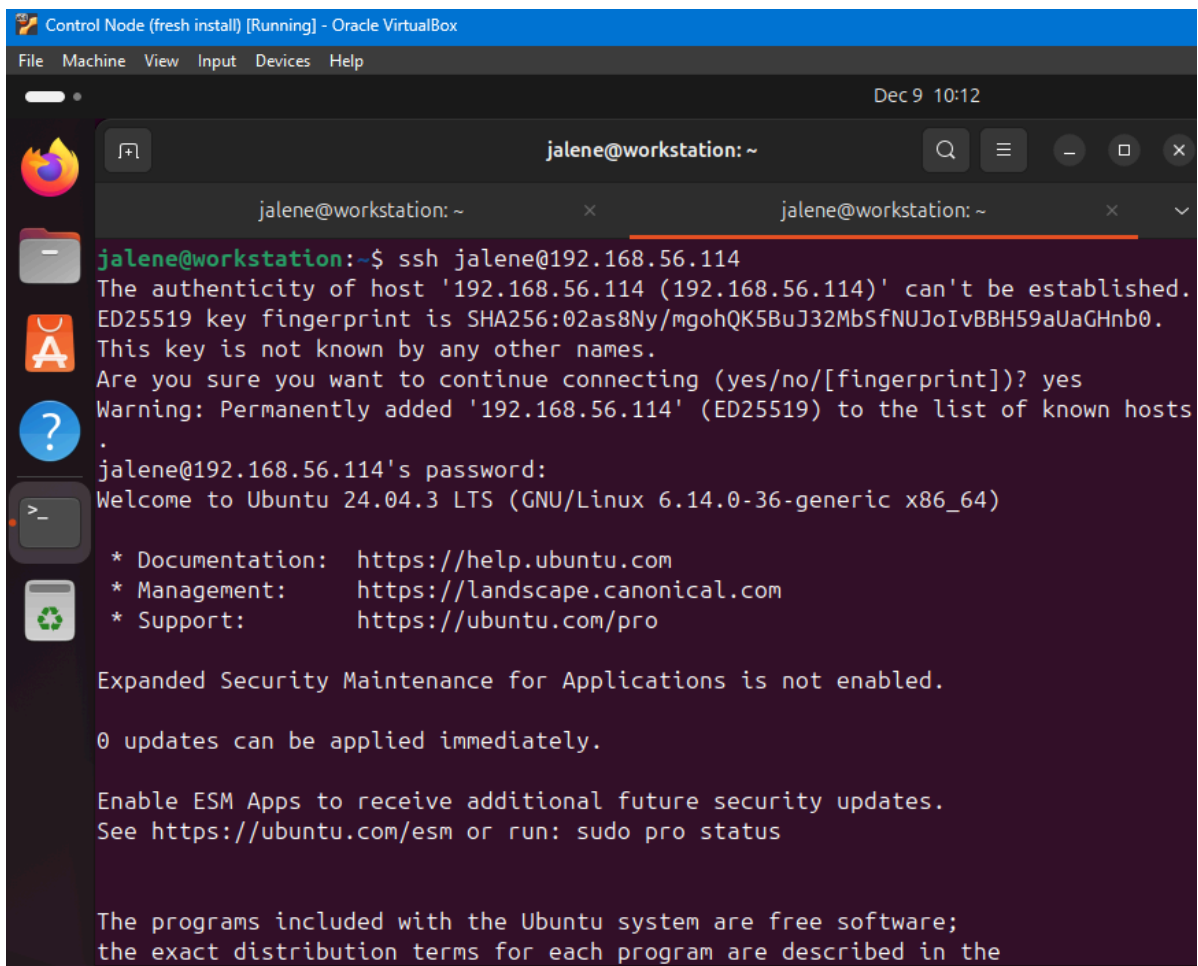
1. On the Local Machine, issue the following commands:

1.1 `ssh username@ip_address_server1` for example, `ssh jvtaylor@192.168.56.120`

1.2 Enter the password for server 1 when prompted

- 1.3 Verify that you are in server 1. The user should be in this format `user@server1`.
For example, `jvtaylor@server1`
2. Logout of Server 1 by issuing the command `control + D`.
3. Do the same for Server 2.
4. Edit the hosts of the Local Machine by issuing the command `sudo nano /etc/hosts`. Below all texts type the following:
 - 4.1 `IP_address server 1` (provide the ip address of server 1 followed by the hostname)
 - 4.2 `IP_address server 2` (provide the ip address of server 2 followed by the hostname)
 - 4.3 Save the file and exit.
5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do `ssh jvtaylor@server1`. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

Local Machine



```
Control Node (fresh install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 9 10:12
jalene@workstation: ~
jalene@workstation: ~
jalene@workstation:~$ ssh jalene@192.168.56.114
The authenticity of host '192.168.56.114 (192.168.56.114)' can't be established.
ED25519 key fingerprint is SHA256:02as8Ny/mgohQK5BuJ32MbSfNUJJoIvBBH59aUaGHnb0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.114' (ED25519) to the list of known hosts
.
jalene@192.168.56.114's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-36-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

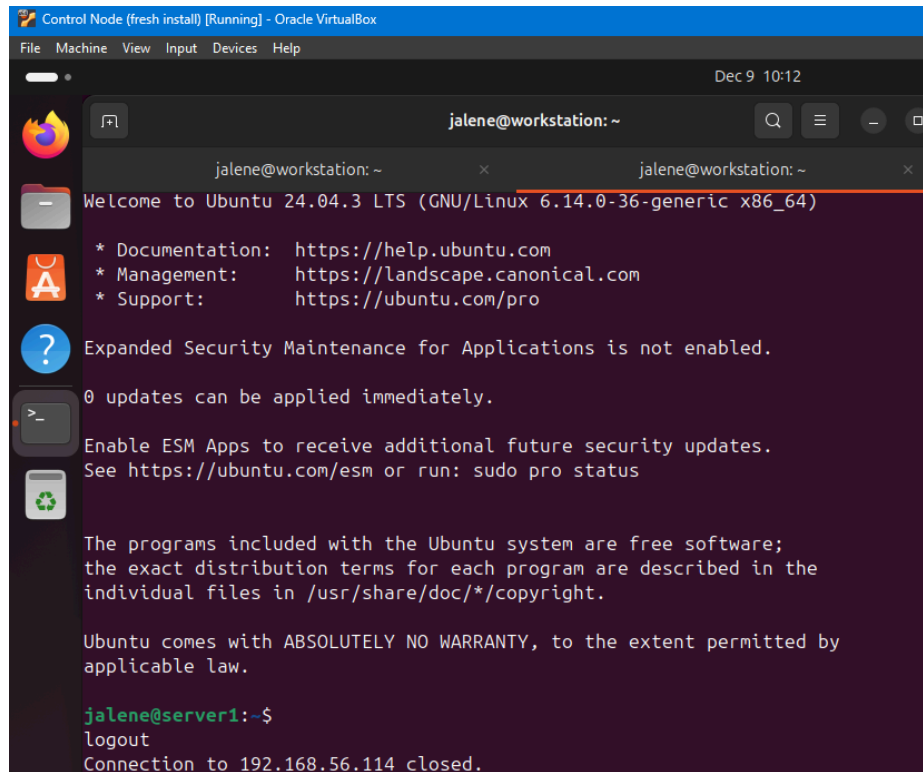
Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
```


Server 1



```
Control Node (fresh install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 9 10:12
jalene@workstation: ~
jalene@workstation: ~
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-36-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

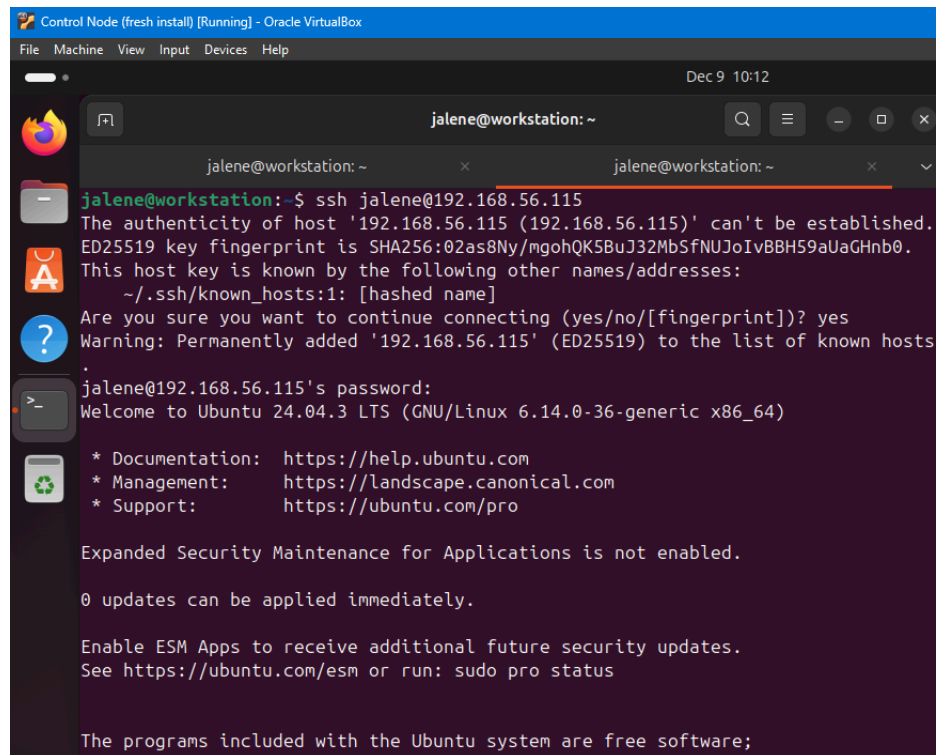
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

jalene@server1:~$
logout
Connection to 192.168.56.114 closed.
```

Server 2



```
Control Node (fresh install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 9 10:12
jalene@workstation: ~
jalene@workstation: ~
jalene@workstation: ~$ ssh jalene@192.168.56.115
The authenticity of host '192.168.56.115 (192.168.56.115)' can't be established.
ED25519 key fingerprint is SHA256:02as8Ny/mgohQK5BuJ32MbSfNUJoIvBBH59aUaGHnb0.
This host key is known by the following other names/addresses:
 ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.115' (ED25519) to the list of known hosts
jalene@192.168.56.115's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-36-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

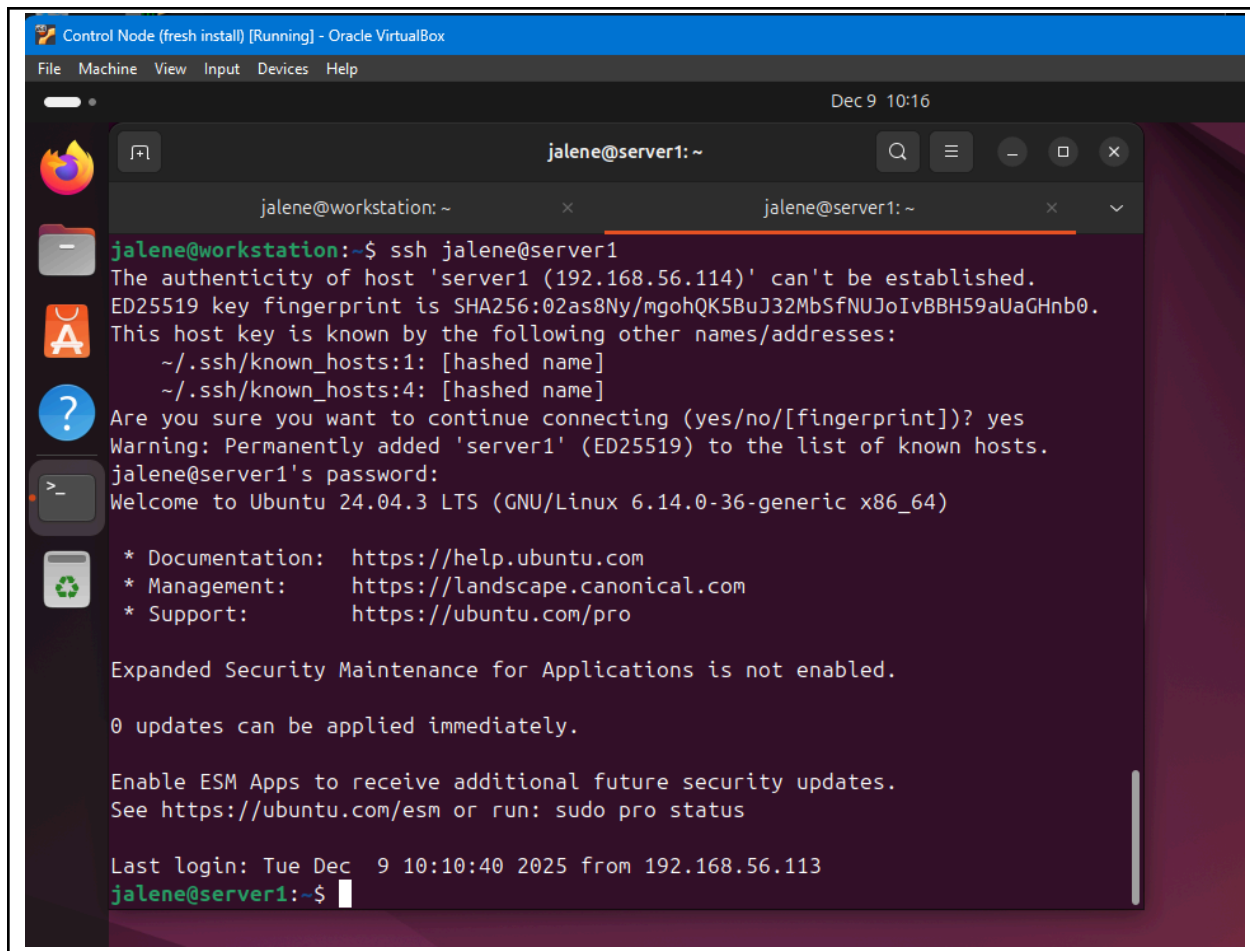
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
```

```
Control Node (fresh install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 9 10:12
jalene@workstation: ~
jalene@workstation: ~
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
jalene@server2:~$
logout
Connection to 192.168.56.115 closed.
jalene@workstation:~$
```

```
Control Node (fresh install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 9 10:15
jalene@workstation: ~
jalene@workstation: ~
GNU nano 7.2 /etc/hosts
127.0.0.1 localhost.localdomain localhost
127.0.0.1 localhost6.localdomain6 localhost6
192.168.56.114 server1
192.168.56.115 server2
# The following lines are desirable for IPv6 capable hosts
::1 localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```



The screenshot shows a terminal window titled "Control Node (fresh install) [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The top right corner shows the date and time "Dec 9 10:26". The terminal content is as follows:

```
jalene@server2: ~  
jalene@workstation: ~  
The authenticity of host 'server2 (192.168.56.115)' can't be established.  
ED25519 key fingerprint is SHA256:02as8Ny/mgohQK5BuJ32MbSfNUJoIvBBH59aUaGHnb0.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: [hashed name]  
  ~/.ssh/known_hosts:4: [hashed name]  
  ~/.ssh/known_hosts:5: [hashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.  
jalene@server2's password:  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-36-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Last login: Tue Dec  9 10:11:31 2025 from 192.168.56.113  
jalene@server2:~$
```

Reflections:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?
I was able to use the hostname instead of the IP address because I added the IP addresses and the manage node's hostnames in /etc/hosts. This lets the system translate the hostnames so the SSH knows where to connect.
2. How secured is SSH?
SSH is indeed very secure because it's designed to encrypt all the data between the client and server. It encrypts all passwords, commands, and files making remote access safe in other networks.